Package 'CrossingTools'

January 23, 2026

Title Tools for Cross Optimization in Plant Breeding Programs

Version 1.0.1

Description Mate allocation is a critical component of plant breeding programs, aiming to generate superior progenies and improved populations over time. While numerous methodologies for mate allocation exist, few software tools offer a unified and scalable framework that integrates these methodologies in a user-friendly manner. 'CrossingTools' addresses this gap by providing a flexible, efficient solution for optimizing mating decisions across a wide range of breeding schemes. The package supports genomic BLUP and selection indices for various variance structures and implements multiple selection criteria to evaluate cross potential, including expected mean, optimal haploid value (OHV), and superior progeny value (SPV). Different methods are provided for calculating additive and dominance segregation variances for both inbred and heterozygous parents, making 'CrossingTools' suitable for a wide range of breeding strategies. After criterion calculation, users can rank potential crosses or apply the built-in optimal crossing selection routine, which balances genetic gain and diversity to support long-term improvement. The package combines R's scripting flexibility with high-performance C++ routines via 'Rcpp' and 'RcppArmadillo', ensuring scalability for large populations with dense marker data.

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Encoding UTF-8

Roxygen list(markdown = TRUE, roclets = c(` `collate", ` `namespace", ` `rd"))

ByteCompile true

RoxygenNote 7.3.3

Depends R (>= 4.0.0)

Imports ggplot2,
ggridges,
Rcpp,
RcppParallel

LinkingTo Rcpp,
RcppArmadillo,
RcppParallel

SystemRequirements C++17

Suggests knitr,
rmarkdown,
testthat (>= 3.0.0)

Config/testthat/edition 3

VignetteBuilder knitr

URL https://github.com/svenernstW/CrossingTools

BugReports https://github.com/svenernstW/CrossingTools/issues

R topics documented:

`calculate_desired_gains` *Calculation of desired gains index (Werner et al.)*

Description

Calculates the desired gains index based on multi-trait BLUPs and a chosen (co)variance matrix option.

Usage

```
calculate_desired_gains(effects, V = NULL, gains = NULL, n.Threads = 4L)
```

Arguments

`effects`	A matrix (n_genotype x n_trait) of multi-trait BLUPs.
`V`	Numeric matrix ((n_genotype*n_trait) x (n_genotype*n_trait)) with the posterior variance var($\tilde{g}$) of the provided effects. If provided, the *marginal* posterior variance (mean of per-genotype trait-blocks from `V`) is used to calculate the index. If not provided an empirical estimate of V as cov(A) is used.
`gains`	Numeric vector (length = n_trait) of desired gains.
`n.Threads`	Number of OpenMP threads.

Value

A list with index values and trait weights.

calculate_optimal_haploid_value

Optimal Haplotypic Value (OHV) for proposed crosses

Description

Computes the OHV for each cross using block-wise local GEBVs. If `haplotype.blocks` is not supplied (or empty), each marker column of `marker.mat` is treated as a separate block.

Usage

```
calculate_optimal_haploid_value(
  crosses,
  haplotype.blocks = NULL,
  marker.mat,
  effects,
  n.Threads = 4L
)
```

Arguments

`crosses`	A numeric matrix or data.frame with two columns (P1, P2), giving indices of parents (rows of `marker.mat`) for each proposed cross.
`haplotype.blocks`	A list of integer vectors. Each vector contains marker indices (columns of `marker.mat`) belonging to that block. If `NULL` or length 0, each marker is used as a separate block.
`marker.mat`	Numeric genotype/marker matrix (individuals x markers).
`effects`	Numeric matrix of marker effects.
`n.Threads`	Integer larger 1. OpenMP threads (if enabled at compile time).

Value

A data.frame with columns `OHV`.

calculate_variances_4W_DH

Segregation variance for DH lines from specified four- or three-way crosses (Allier)

Description

Calculate expected genomic estimated breeding values (EGBV), segregation variance, and superior progeny value (SPV) for doubled haploid (DH) lines derived after `t` rounds of random mating.

Usage

```
calculate_variances_4W_DH(
  crosses,
  genetic.map,
  marker.mat,
  effects,
  t,
  intensity,
  covariance = FALSE,
  weights = NULL,
  n.Threads = 4L
)
```

Arguments

crosses	A data.frame or matrix (n_crosses x 4) of parental indices (row indices into marker.mat) specifying four- or three-way crosses. For three-way crosses, the first two crossing partners are the same. You can also compute the variance for two heterozygous individuals by passing haplotypes in marker.mat; then indices refer to haplotypes.
genetic.map	A data.frame with columns: • site: integer marker index (1..ncol(marker.mat)) • chr: chromosome identifier • pos: position on the chromosome in morgan All markers in marker.mat must appear in genetic.map$site.
marker.mat	Numeric marker matrix (genotypes/haplotypes in rows, markers in columns) with entries in c(0, 2) corresponding to c("AA", "BB"). For heterozygotes, store haplotypes in marker.mat (each individual uses two rows).
effects	Numeric matrix of marker effects (markers in rows, traits in columns). Must have nrow(effects) == ncol(marker.mat).
t	Integer. Number of random-mating generations before DH creation.
intensity	Numeric. Standardized selection differential.
covariance	Logical. If TRUE, also compute the segregation covariance matrix between all supplied traits.
weights	Numeric vector of length ncol(effects) with fixed trait weights. If supplied the function calculates index values for each cross. Only works if covariance = TRUE; otherwise ignored.
n.Threads	Integer (default 4). Number of OpenMP threads (if enabled at compile time).

Value

If covariance = TRUE, a list with element cross_values (a data.frame) whose columns are named EGBV1, var1, SPV1, EGBV2, ... and a list with element covariances with segregation covariance matrix for each cross. If covariance = TRUE, a list with element cross_values (a data.frame) whose columns are named EGBV1, var1, SPV1, EGBV2, ..., element index (a data.frame) with EGBV_DG, varDG, SPVDG for the index and a list with element covariances with segregation covariance matrix for each cross. If covariance = FALSE, a data.frame with the same columns EGBV1, var1, SPV1, EGBV2, ...

Examples

```
## Not run:
out <- calculate_variances_DH(
  crosses = matrix(c(1,2, 3,4), ncol = 4, byrow = TRUE),
 genetic.map = data.frame(site = 1:ncol(marker.mat), chr = 1, pos = seq_len(ncol(marker.mat))),
  marker.mat = marker.mat,
  effects = matrix(rnorm(ncol(marker.mat)), ncol = 1),
  t = 0,
  intensity = 1.0,
  covariance = FALSE
)

## End(Not run)
```

`calculate_variances_4W_RIL` *Segregation variance for RIL from specified four- or three-way crosses (Allier)*

Description

Calculate expected genomic estimated breeding values (EGBV), segregation variance, and superior progeny value (SPV) for recombinant inbred lines(RIL) derived after `t` rounds of random mating.

Usage

```
calculate_variances_4W_RIL(
  crosses,
  genetic.map,
  marker.mat,
  effects,
  t,
  intensity,
  covariance = FALSE,
  weights = NULL,
  n.Threads = 4L
)
```

Arguments

`crosses`	A data.frame or matrix (n_crosses x 4) of parental indices (row indices into `M`) specifying four- or three-way crosses. For three-way crosses, the first two crossing partners are the same. You can also compute the variance for two heterozygous individuals by passing haplotypes in `M`; then indices refer to haplotypes.
`genetic.map`	A data.frame with columns: • `site`: integer marker index (1..ncol(M)) • `chr`: chromosome identifier • `pos`: position on the chromosome in morgan All markers in `M` must appear in `genetic.map$site`.

marker.mat	Numeric marker matrix (genotypes/haplotypes in rows, markers in columns) with entries in c(0, 2) corresponding to c("AA", "BB"). For heterozygotes, store haplotypes in M (each individual uses two rows).
effects	Numeric matrix of marker effects (markers in rows, traits in columns). Must have nrow(U) == ncol(M).
t	Integer. Number of random-mating generations before RIL creation.
intensity	Numeric. Standardized selection differential.
covariance	Logical. If TRUE, also compute the segregation covariance matrix between all supplied traits.
weights	Numeric vector of length ncol(U) with fixed trait weights. If supplied the function calculates index values for each cross. Only works if covariance = TRUE; otherwise ignored.
n.Threads	Integer (default 4). Number of OpenMP threads (if enabled at compile time).

Value

If covariance = TRUE, a list with element cross_values (a data.frame) whose columns are named EGBV1, var1, SPV1, EGBV2, ... and a list with element covariances with segregation covariance matrix for each cross. If covariance = TRUE, a list with element cross_values (a data.frame) whose columns are named EGBV1, var1, SPV1, EGBV2, ..., element index (a data.frame) with EGBV_DG, varDG, SPVDG for the desired gains index and a list with element covariances with segregation covariance matrix for each cross. If covariance = FALSE, a data.frame with the same columns EGBV1, var1, SPV1, EGBV2, ...

Examples

```
## Not run:
out <- calculate_variances_RIL(
  crosses = matrix(c(1,2, 3,4), ncol = 4, byrow = TRUE),
  genetic.map = data.frame(site = 1:ncol(M), chr = 1, pos = seq_len(ncol(M))),
  marker.mat = M,
  effects = matrix(rnorm(ncol(M)), ncol = 1),
  t = 0,
  intensity = 1.0,
  covariance = FALSE
)

## End(Not run)
```

---

calculate_variances_DH

Segregation variance for DH lines from specified crosses

Description

Calculate expected genomic estimated breeding values (EGBV), segregation variance, and superior progeny value (SPV) for doubled haploid (DH) lines derived after t rounds of random mating.

Usage

```
calculate_variances_DH(
  crosses,
  genetic.map,
  marker.mat,
  effects,
  t,
  intensity,
  covariance = FALSE,
  weights = NULL,
  method = "osthushenrich",
  n.Threads = 4L
)
```

Arguments

`crosses`	A data.frame or matrix (n_crosses x 2) of parental indices (row indices into `marker.mat`) specifying the crosses.
`genetic.map`	A data.frame with columns: • `site`: integer marker index (1..ncol(marker.mat)) • `chr`: chromosome identifier (numeric) • `pos`: position on the chromosome in morgan All markers in `marker.mat` must appear in `genetic.map$site`.
`marker.mat`	Numeric marker matrix (markers in columns, genotypes in rows) with entries in `c(0, 2)` corresponding to `c("AA", "BB")`.
`effects`	Numeric matrix of marker effects (markers in rows, traits in columns). Must have `nrow(effects) == ncol(marker.mat)`.
`t`	Integer. Number of random-mating generations before DH creation.
`intensity`	Double. Standardized selection differential.
`covariance`	Logical. If `TRUE`, also compute the segregation covariance matrix between all supplied traits.
`weights`	Numeric vector of length `ncol(effects)` with fixed trait weights. If supplied the function calculates index values for each cross. Only works if `covariance = TRUE`; otherwise ignored.
`method`	Character. Which method to use; one of `c("osthushenrich", "lehermeier")`.
`n.Threads`	Integer (default 4). Number of OpenMP threads (if enabled at compile time).

Value

If `covariance = TRUE`, a list with element `cross_values` (a data.frame) whose columns are named `EGBV1, var1, SPV1, EGBV2, ...` and a list with element `covariances`. If `covariance = TRUE` and `calculate.index = TRUE`, additionally an element `index` (a data.frame) with `IDG_A, VARIDG_A, SPVIDG_A`. If `covariance = FALSE`, a data.frame with the same columns `EGBV1, var1, SPV1, EGBV2, ....`

`calculate_variances_F1` *Additive and dominance segregation variance of families from specified crosses (Wolfe)*

Description

Calculate expected genomic estimated breeding value (EGBV), expected total genetic value (ETGV), additive and dominance segregation variance, superior progeny value with additive variance (SPV), and superior progeny value including dominance variance (TSPV) for F1 families.

Usage

```
calculate_variances_F1(
  crosses,
  genetic.map,
  hap.mat1,
  hap.mat2,
  effects.A,
  effects.D,
  intensity,
  covariance,
  weights = NULL,
  n.Threads = 4L
)
```

Arguments

`crosses`	A data.frame or matrix (n_crosses x 2) of parental indices (row indices into `hap.mat1`/`hap.mat2`) specifying the crosses.
`genetic.map`	A data.frame with columns: • `site`: integer marker index (1..ncol(hap.mat1)) • `chr`: chromosome identifier (numeric or factor) • `pos`: position on the chromosome in morgan All markers in `hap.mat1`/`hap.mat2` must appear in `genetic.map$site`.
`hap.mat1`	Numeric haplotype matrix (individuals x markers) for the first haplotype, entries in `c(0, 1)` for `c("A", "B")`.
`hap.mat2`	Numeric haplotype matrix (individuals x markers) for the second haplotype, entries in `c(0, 1)` for `c("A", "B")`.
`effects.A`	Numeric matrix of additive marker effects (markers in rows, traits in columns). Must have `nrow(effects.A) == ncol(hap.mat1)`.
`effects.D`	Numeric matrix of dominance marker effects (markers in rows, traits in columns). Must have `nrow(effects.D) == ncol(hap.mat1)` and `ncol(effects.D) == ncol(effects.A)`.
`intensity`	Double. Standardized selection differential (used for SPV).
`covariance`	Logical. If `TRUE`, also compute the segregation covariance matrix between all supplied traits.

weights	Numeric vector of length ncol(effects.A) with fixed trait weights. If supplied the function calculates index values for each cross. Only works if covariance = TRUE; otherwise ignored.
n.Threads	Integer (default 4). Number of OpenMP threads (if enabled at compile time).

Value

If covariance = TRUE, a list with element cross_values (a data.frame) whose columns are named EGBV1, ETGV1, varA1, SPV1, varD1, TSPV1, EGBV2, ... and covariance matrices per cross. If covariance = TRUE and calculate.index = TRUE, additionally an element index (a data.frame) with IDG_A, VARIDG_A, SPVIDG_A, IDG_AD, VARIDG_AD, SPVIDG_AD. If covariance = FALSE, a data.frame with the same columns.

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calculate_variances_RIL

Segregation variance for recombinant inbred lines from specified crosses

---

Description

Calculate expected genomic estimated breeding values (EGBV), segregation variance, and superior progeny value (SPV) for recombinant inbred lines (RIL) derived after t rounds of random mating.

Usage

```
calculate_variances_RIL(
  crosses,
  genetic.map,
  marker.mat,
  effects,
  t,
  intensity,
  covariance = FALSE,
  weights = NULL,
  method = "osthushenrich",
  n.Threads = 4L
)
```

Arguments

crosses	A data.frame or matrix (n_crosses x 2) of parental indices (row indices into M) specifying the crosses.
genetic.map	A data.frame with columns: • site: integer marker index (1..ncol(M)) • chr: chromosome identifier (numeric) • pos: position on the chromosome in morgan All markers in M must appear in genetic.map$site.
marker.mat	Numeric marker matrix (markers in columns, genotypes in rows) with entries in c(0, 2) corresponding to c("AA", "BB").

effects	Numeric matrix of marker effects (markers in rows, traits in columns). Must have nrow(U) == ncol(M).
t	Integer. Number of random-mating generations before RIL creation.
intensity	Double. Standardized selection differential.
covariance	Logical. If TRUE, also compute the segregation covariance matrix between all supplied traits.
weights	Numeric vector of length ncol(U) with fixed trait weights. If supplied the function calculates index values for each cross. Only works if covariance = TRUE; otherwise ignored.
method	Character. Which method to use; one of c("osthushenrich", "lehermeier").
n.Threads	Integer (default 4). Number of OpenMP threads (if enabled at compile time).

Value

If covariance = TRUE, a list with element cross_values (a data.frame) whose columns are named EGBV1, var1, SPV1, EGBV2, ... and a list with element covariances. If covariance = TRUE and calculate.index = TRUE, additionally an element index (a data.frame) with IDG_A, VARIDG_A, SPVIDG_A. If covariance = FALSE, a data.frame with the same columns EGBV1, var1, SPV1, EGBV2, ...

---

evaluate_cross_plan *Evaluate a cross plan by averaging value metrics across all crosses*

Description

Evaluate a cross plan by averaging value metrics across all crosses

Usage

```
evaluate_cross_plan(cross_plan, cross_param)
```

Arguments

cross_plan	A data.frame with two columns as produced by create_cross_plan().
cross_param	Either a data.frame (when covariance = FALSE) or a list with element $cross_values (data.frame) when covariance = TRUE, produced by calculate_*.

Value

A single-row data.frame containing the mean across all crosses for each available value metric per trait (among EGBV, ETGV, SPV, SPTV, TSPV), with columns named mean_<METRIC><trait_index>, plus n_crosses.

get_marker_effects *Backsolve marker effects (mu) from individual effects (g)*

Description

Computes marker effects $\mu = (scalingFactor * M' * G^{-1}) * g$.

Usage

```
get_marker_effects(M, G, effects, scaling.factor, tol = 1e-10, n.Threads = 4L)
```

Arguments

G	Numeric matrix (n x n): relationship matrix among individuals.
effects	Numeric (n x k) or length-n vector: individual effects (one or more traits).
scaling.factor	Numeric scalar used to scale the GRM.
tol	Numeric scalar tolerance for near-singularity checks in G. Default: 1e-10.
n.Threads	Integer >= 1; OpenMP threads (if enabled at compile time). Default: 4L.
marker.mat	Numeric matrix (n x p): marker/genotype matrix (individuals in rows, markers in columns).

Value

A numeric p x k matrix of marker effects (mu_matrix).

make_cross_plan *Creation of a plan of all potential crosses for CrossingTools*

Description

Creation of a plan of all potential crosses for CrossingTools

Usage

```
make_cross_plan(
  candidates = NULL,
  male_candidates = NULL,
  female_candidates = NULL
)
```

Arguments

candidates	A vector of integers that identify each genotype. Should be consistent with genotypic data. Use this function if there are no sexes.
male_candidates	A vector of integers that identify each male genotype. Ignored if candidates is supplied. Should be consistent with genotypic data.
female_candidates	A vector of integers that identify each female genotype. Ignored if candidates is supplied. Should be consistent with genotypic data.

Value

A two-column data.frame with the the combinations of all supplied individuals. If male and female candidates are supplied, the first column corresponds to males, and the second to females

---

`optimal_cross_selection` *Optimal cross selection via a genetic algorithm (with optional fixed crosses)*

Description

Uses a simple genetic algorithm (GA) to choose `ncrosses` parent pairs that balance an average criterion (utility `u`) and genomic similarity according to a target trade-off angle. Fixed crosses (if any) are forced into the final solution, and the GA fills the remaining slots from `crosses`.

Usage

```
optimal_cross_selection(
  crosses,
  fixed.crosses = NULL,
  ncrosses,
  target.angle,
  u,
  u.fixed = NULL,
  G,
  return.params = FALSE,
  params = list(),
  n.Threads = 4L
)
```

Arguments

`crosses`	integer matrix (K x 2). Candidate *variable* crosses (1-based indices referring to rows/columns of `G`). Must have exactly two columns (P1, P2); selfings are not allowed.
`fixed.crosses`	integer matrix (F x 2) or `NULL`. Crosses that must be included in the final plan. Can have 0 rows. Must use the same indexing as `crosses`. Selfings are not allowed.
`ncrosses`	integer. Total number of crosses in the final plan (variable + fixed). Must be $\geq$ number of rows in `fixed.crosses`.
`target.angle`	numeric (radians). Target trade-off angle between the average criterion and the similarity objective. Smaller angles emphasize utility `u`; larger angles emphasize similarity control.
`u`	numeric vector (length = nrow(crosses)). Utility (average criterion) for each candidate cross in `crosses`.
`u.fixed`	numeric vector (length = nrow(fixed.crosses)) or `NULL`. Utility for each fixed cross; if `fixed.crosses` has 0 rows, this can be length-0.

`G` numeric square matrix (nInd x nInd). Genomic relationship matrix used to evaluate similarity/diversity among parents; indices in `crosses`/`fixed.crosses` must lie in `1..nrow(G)`.

`return.params` logical. If `TRUE`, return additional information from the GA.

`params` A list of parameters for the Genetic Algorithm optimization:

- `propability.mutate` numeric in $[0, 1]$. Probability that a candidate solution mutates in the GA.
- `n.mutate` integer ≥ 0. Number of point mutations to apply when a mutation occurs.
- `n.select` integer ≥ 2. Number of candidate solutions selected per generation (selection pool size).
- `n.pop` integer ≥ 2. Population size (number of candidate solutions) per generation.
- `max.generation` integer ≥ 1. Maximum number of GA generations.
- `max.iteration` integer ≥ 1. Maximum number of iterations without improvement.
- `angle.penalty` numeric ≥ 0. Penalty applied to solutions that deviate from `target.angle`.

`n.Threads` integer ≥ 1. Number of OpenMP threads to use.

Value

If `return.params = FALSE`**:** A data.frame with columns `parent1` and `parent2` describing the selected cross plan.

If `return.params = TRUE`**:** A list with elements:

- `crossPlan` A data.frame with columns `parent1` and `parent2` describing the selected cross plan.
- `uMax` Numeric scalar: maximum attainable average criterion when optimizing only `u`.
- `uMin` Numeric scalar: minimum attainable average criterion when optimizing only similarity.
- `simMax` Numeric scalar: similarity corresponding to `uMax`.
- `simMin` Numeric scalar: similarity corresponding to `uMin`.
- `uBest` Numeric scalar: average criterion achieved by the optimized cross plan.
- `simBest` Numeric scalar: similarity of the optimized cross plan.
- `angleBest` Numeric scalar (radians): angle of the optimized solution relative to the target trade-off.
- `lenBest` Numeric scalar: length (magnitude) of the optimized solution vector in the objective space.

---

plot_cross_plan *Plot cross plan ridge distributions per trait*

Description

Simulates per-cross distributions using EGBV as mean and the sum of all non-dominance variance columns (i.e., all var* except varD) as the variance. Draws ridgeline densities, overlays EGBV (blue circles) and SPV (red diamonds, if present), and adds a dotted vertical line at the per-trait average EGBV. One panel per trait (3 columns), each with its own x-axis scale.

Usage

```
plot_cross_plan(cross_plan, cross_param, traits = NULL)
```

Arguments

cross_plan	data.frame: two columns from create_cross_plan()
cross_param	data.frame or list with $cross_values from calculate_variances*()
traits	integer vector of trait indices to plot, or NULL for all

Value

ggplot object

Index